

Find $f(-3)$, $f(0)$, $f(2)$

$$42. \quad f(x) = \begin{cases} 3 - \frac{1}{2}x & \text{if } x < 2 \\ 2x - 5 & \text{if } x \geq 2 \end{cases}$$

$f(x): x^2+1$ odd, even, neither?

$$f(-x) = (-x)^2+1 = x^2+1 = f(x)$$

DAY 2: DOMAIN, RANGE, & INVERSE FUNCTIONS

1) TODAY'S GOAL

By the end of Today you should be able to answer:

- What is a domain?
- What is range?
- How do I find domain & range?
- What does one to one mean?
- What is an inverse function & how do I find it?

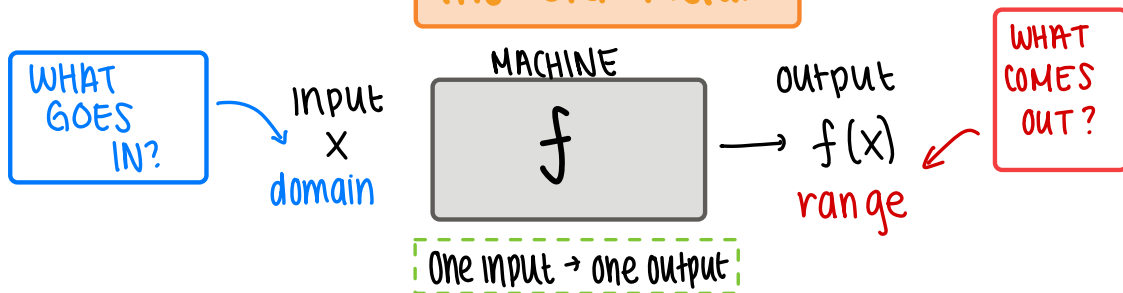
2) REVIEW from LAST

We introduced functions as rules that take an input & give exactly one output.



This idea is the foundation for everything we will do today!

THE BIG PICTURE



DOMAIN (inputs)

The domain is the set of all allowed inputs.

- x values
- Lives horizontally on a graph (read left $\rightarrow$ right)
- Everything the machine accepts

RANGE (outputs)

The range is the set of all possible outputs.

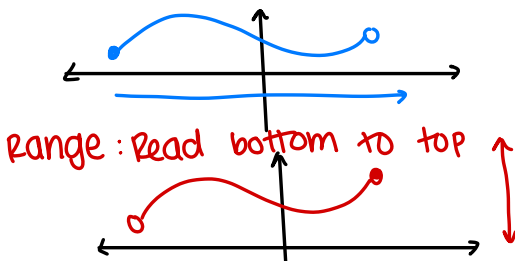
- y -values
- Lives vertically on a graph (read bottom to top)
- Everything the machine can produce

REAL WORLD EXAMPLES

If $f(t)$ gives the temperature ($^{\circ}\text{F}$) @ time t (in hours) then:

- **INPUT**: t (time)
- **DOMAIN**: times measured
- **RANGE**: set of temp. occurred

Domain: read left $\rightarrow$ right



IMPORTANT

when a domain is not given we usually start by:

- 1) Assuming all real numbers are allowed ($\mathbb{R}$)
- 2) Then removing any values that break the formula or don't make sense

1) FINDING DOMAIN FROM A FORMULA

STRATEGY: HOW TO FIND DOMAIN

- ① Start with ALL real numbers
 $D = (-\infty, \infty)$
- ② Find input values that break the function / make the output not exist / not make sense
- ③ Remove those values from the set
what's left is the domain!

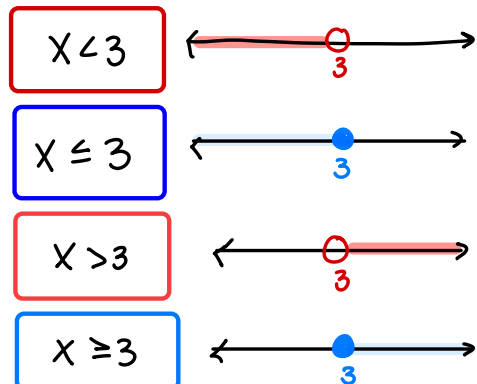
NOTATION DECODER

- () parenthesis = NOT included (open endpt) | • $x \leq a$ & $x \geq a \rightarrow$ closed endpoints (use brackets)
- [] brackets = INCLUDE (closed endpt) | • ∞ & $-\infty$ are ALWAYS in ()
- $\circ$ open circle = not included | • The union symbol $\cup$ means "combine pieces" or OR
- $\bullet$ closed circle = included
- $x < a$ & $x > a \rightarrow$ open endpoints (use parenthesis)

COMMON RESTRICTIONS

- 1) Denominator cannot equal 0
($x \neq$ value that makes denom 0).
- 2) Inside an EVEN root must be greater than or equal to 0 (≥ 0)
- 3) Inside a LOGARITHM must be greater than 0 (> 0)
- 4) Tan is undefined when $\cos x = 0$
(tan x undefined @ $x = \frac{\pi}{2} + k\pi$)
- 5) If there are MULTIPLE restrictions ALL must be true

INEQUALITY $\sim$ NUMBER LINE



EXAMPLES

① Polynomial $f(x) = x^3 - 4x + 7$	Polynomials are defined for all real numbers ($\mathbb{R}$)	$D = (-\infty, \infty)$
② Rational $g(x) = \frac{1}{x-3}$	Denominator cannot be $= 0$ $x-3=0 \rightarrow x \neq 3$	$D = (-\infty, 3) \cup (3, \infty)$
③ Square Root $h(x) = \sqrt{5-2x}$	Inside root ≥ 0 $5-2x \geq 0 \rightarrow -2x \geq -5 \rightarrow x \leq \frac{5}{2}$	$D = (-\infty, \frac{5}{2}]$
④ Logarithm $p(x) = \ln(x+4)$	Inside log > 0 $x+4 > 0 \rightarrow x > -4$	$D = (-4, \infty)$
⑤ MULTIPLE $a(x) = \frac{\sqrt{x+1}}{x+2}$	Root: $x+1 \geq 0 \rightarrow x \geq -1$ Denom: $x+2 \neq 0 \rightarrow x \neq -2$ Both must be true	$D = [-1, 2) \cup (2, \infty)$

ON A GRAPH

open circles on a graph go w/ ()

closed circles on a graph go with []

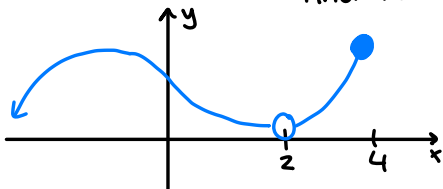
PRACTICE: Find domain $q(x) = \frac{1}{x^2-4}$

HOW TO READ DOMAIN FROM A GRAPH

- Follow the graph from the **left most** x value to the **right most** x-value.
- ARROWS mean the graph continues forever in that direction
- open circle (○) means not included (use parenthesis)
- closed circle (●) means that x-value **is included** (brackets)
- if there is a **gap or a hole** use the union symbol $\cup$

WORKED EX:

Find Domain:



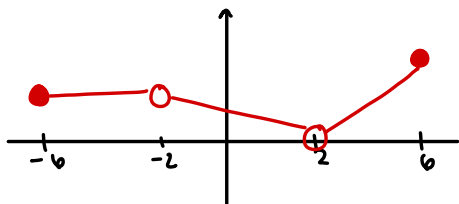
↳ graph continues forever $\leftarrow (-\infty)$

↳ open circle @ $x=2 \rightarrow 2$ not included

↳ closed circle @ $x=4 \rightarrow 4$ is included

$$D = (-\infty, 2) \cup (2, 4]$$

WORKED EX: Find Domain.



closed $x = -6 \rightarrow$ include -6

open $x = -2 \rightarrow$ don't include -2

open $x = 2 \rightarrow$ don't include 2

closed $x = 6 \rightarrow$ include 6

$$D = [-6, -2) \cup (-2, 2) \cup (2, 6]$$

PRACTICE 1) $f(x) = 3x^2 - 1$ $g(x) = \frac{1}{x+5}$ $h(x) = \sqrt{x-7}$ $p(x) = \ln(2x-1)$

HOW TO FIND RANGE

- 1 Read it from the graph
- 2 Use known graph shapes
- 3 Use algebra & inequalities
- 4 Think about the smallest and largest possible outputs

Ex: FORMULA

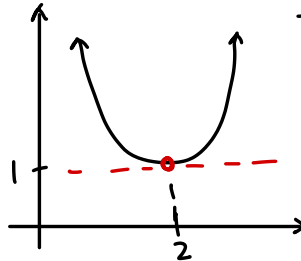
$$f(x) = (x-2)^2 + 1$$

$$(x-2)^2 \geq 0 \quad \text{always true}$$

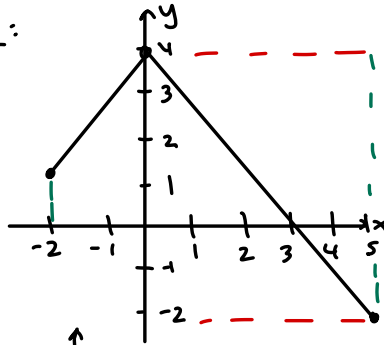
$$(x-2)^2 + 1 \geq 1 \quad \text{add 1}$$

Therefore, $f(x) \geq 1$

Range $[1, \infty)$



Ex:



$$D: [-2, 5]$$

$$R: [-1, 4]$$

Ex: Square root formula

$$q(x) = \sqrt{x-1} + 2$$

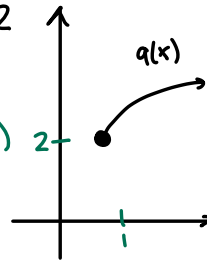
$$\sqrt{x-1} \geq 0 \quad \text{square roots} \geq 0$$

$$\sqrt{x-1} + 2 \geq 2 \quad (\text{add 2})$$

Therefore $q(x) \geq 2$

$$q(x) \geq 2$$

Range $[2, \infty)$



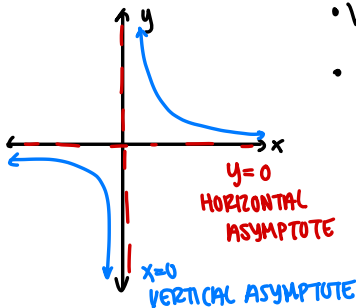
You try: what is range of $y = (x+1)^2$

$$f(x) = x^2 + 4$$

$$g(x) = -x^2 + 1$$

$$h(x) = \sqrt{x} - 5$$

Ex: $f(x) = 1/x$



- Vertical asymptote $x=0$
- Horizontal asymptote $y=0$

$$D: (-\infty, 0) \cup (0, \infty)$$

$$R: (-\infty, 0) \cup (0, \infty)$$

We can never have $y=0$ so 0 not in range

Ex $h(x) = 3 - \sqrt{x+1}$

$$\sqrt{x+1} \geq 0 \quad \text{square root is never negative}$$

$$\text{Therefore } -\sqrt{x+1} \leq 0$$

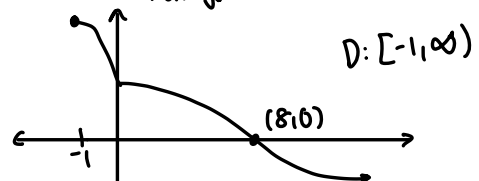
$$\text{So, } 3 - \sqrt{x+1} \leq 3$$

$$\text{At } x=-1, h(-1) = 3 - \sqrt{-1+1} = 3$$

largest value = 3

$$\text{Range} = (-\infty, 3]$$

$(-1, 3)$



$$D: [-1, \infty)$$

INVERSE FUNCTIONS: THE BIG IDEA

Definition: An inverse function undoes what another function does. It takes the output of a function & gives back original input

THE BIG PICTURE FORWARD VS BACKWARD

Original function		Inverse function	
x	$\rightarrow f(x)$	$f(x)$	$\rightarrow x$
Input	output	Inputs	outputs

DOMAIN & RANGE SWAP

If f has

domain A range B

Then f^{-1} has

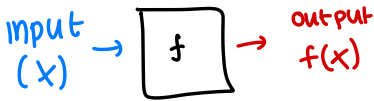
domain B range A

WHY DOES DOMAIN & RANGE MATTER?

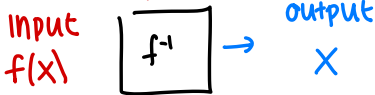
- A function can only accept inputs from domain
- The outputs it creates form its range.
- The inverses take those outputs & uses them as inputs

so the range of the original must become the domain of the inverse.

Forward (f)



Backward (f^{-1})



Ex: If $f(2) = 5$ then $f^{-1}(5) = 2$
Inverse takes 5 (an output) and returns 2 (o.g input)

Practice: Given $D: [1,4]$ $R: [-2,7]$
 $D(f^{-1})$:
 $R(f^{-1})$.

ONE TO ONE

A function f is one to one (injective) if different inputs give different outputs

$$a \neq b \Rightarrow f(a) \neq f(b)$$

ex:

x	1	2	3
y	4	4	7

Is it a function? Yes
Is it one to one? No
 $1 \neq 2$ $f(1) = f(2) = 4$

ONE TO ONE $\rightarrow$ INVERSES

If a function is NOT one to one then the inverse would give one input 2 outputs

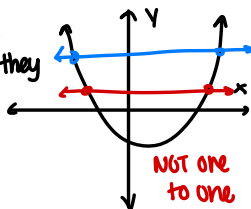


means the "inverses" would not be a function (b/c a function must have exactly one output for each inputs)

CHECKING w/ GRAPHS: HORIZONTAL LINE TEST

A function is one to one if every horizontal line intersects the graph at most once

We draw horizontal lines
and count how many times they
hit the graph



You try: Is $f(x) = x^2 + x$
one to one?

Finding an Inverse Algebr.

Example: $f(x) = 2x - 3$

Steps

- 1) write $y = f(x)$
- 2) Switch x & y
- 3) solve for y
- 4) Rename y as $f^{-1}(x)$
- 5) check domain & range

1) $y = 2x - 3$

2) $x = 2y - 3$

3) $x + 3 = 2y$

4) $y = \frac{x+3}{2}$

5) $f^{-1}(x) = \frac{1}{2}(x+3)$

CHECK

1. If $f(4) = 5$

then $f^{-1}(5) = 4$

1. $f(4) = 2(4) - 3 = 5$

1. $f^{-1}(5) = \frac{1}{2}(5+3) = 4$

Inverse Notation

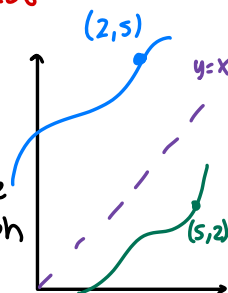
$$\left. \begin{array}{l} f^{-1}(y) = x \\ f^{-1}(x) = y \end{array} \right\} f^{-1} \neq \frac{1}{f(x)}$$
$$\left. \begin{array}{l} f(x) = y \\ f(y) = x \end{array} \right\}$$

GRAPH OF AN INVERSE

• If (a, b) is on the graph of f then (b, a) is on the graph of f^{-1}

• The graph of f^{-1} is the **REFLECTION** of the graph of f across line $y = x$

* swap coordinates

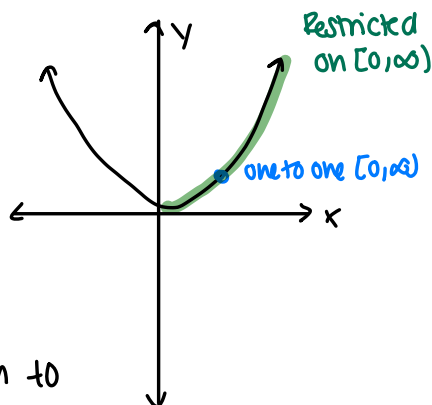


Note: $f(x) = x^2$

• On all real numbers: $f(x) = x^2$ is **NOT** one to one b/c $f(2) = f(-2) = 4$

• Restrict domain to $x \geq 0$ Then $f(x) = x^2$ is one to one on $[0, \infty)$

• Inverse: $f(x) = x^2, x \geq 0 \Rightarrow f^{-1}(x) = \sqrt{x} \quad x \geq 0$



* **NOTE:** we can restrict the domain of a function to make it one-to-one.

Example: $f(x) = \sqrt{x-1} + 2$

ORIGINAL FUNCTIONS

- Domain: $[1, \infty)$
- Range: $[2, \infty)$

INVERSE FUNCTIONS

- Domain: $[2, \infty)$
- Range: $[1, \infty)$

FIND INVERSE

1) $y = \sqrt{x-1} + 2$

2) $x = \sqrt{y-1} + 2$

3) $x-2 = \sqrt{y-1}$

4) $(x-2)^2 = y-1$

5) $(x-2)^2 + 1 = y$

6) $y = (x-2)^2 + 1$

$f^{-1}(x) = (x-2)^2 + 1$

Note: the inverse
graph is NOT the full parabola
its domain restricted to $x \geq 2$

